

Project Changes:

The main research question is still the same: do stocks held by the same large institutional investors tend to move together more during periods of market stress? I have made a few methodological changes for clarity.

First, instead of directly analyzing the bipartite firm-institution network, I am now constructing a projected firm-firm network. Institutional ownership is first represented as a bipartite network between firms and institutions using 13F filings from WRDS. This was then projected into a weighted firm level network using matrix multiplication, where the weight between two firms reflects their ownership overlap. I think this made the empirical design more directly aligned with the research question, and concerning comovement between firms.

Second, I finalized the definition of market stress at the daily level. Stress days are defined as days in the top 10% of realized S&P 500 volatility, calculated from daily returns retrieved through the FRED API. This provides more observations for correlation estimation and preserves variation across quarters.

Third, I am now computing quarterly pairwise return correlations during stress periods and regressing these correlations on quarterly ownership overlap. The regression results showed a positive and statistically significant relationship between ownership overlap and stress period return correlation.

No changes were made to the data sources. Yahoo Finance is still used to construct the firm sample, FRED provides S&P 500 data for stress identification, and WRDS provides daily returns and 13F institutional ownership data.

Data Analysis and Visualizations Plan:

I constructed quarterly firm-firm ownership overlap measures and computed pairwise return correlations during stress periods. An OLS regression of stress period correlation on ownership overlap yielded a positive and statistically significant coefficient ($\beta = 12.46$, $p < 0.001$, $R^2 = 0.027$). This suggests that firms sharing greater institutional ownership exhibit stronger comovement during market stress.

Specific Data Analysis Decision:

Instead of counting the number of shared institutional investors, I construct a weighted overlap measure using matrix multiplication ($W W'$), where W is the firm-by-institution ownership percentage matrix. This captures whether two firms share institutions as well as the magnitude of overlapping ownership. So now larger common holdings contribute more to the network weight than small incidental overlaps.

Additionally, I applied a minimum ownership threshold of 0.1% when constructing the ownership network. This filtered out extremely small institutional positions that are unlikely to reflect meaningful economic influence and reduces noise in the overlap matrix.

Visualizations:

1. A network graph of the firm-firm ownership network, with node size proportional to weighted degree centrality and edge thickness proportional to ownership overlap
2. A histogram of pairwise stress-period correlations
3. A scatter plot of ownership overlap versus stress correlation with a fitted regression line

Ownership overlap values are small (typically between 0.002 and 0.02), which compresses the x axis and makes interpretation less intuitive. I might rescale ownership overlap into percentage terms to improve interpretability. I also plan to aggregate results at the quarter level and facet the scatter plot by quarter to examine whether the relationship is stronger in certain stress episodes.

Focus Area

The focus area of this project is network analysis and financial econ. Specifically, the project examines how institutional ownership networks are related to stock return comovement during periods of market stress.

Link Between Empirical Strategy and Research Question:

- The dependent variable is pairwise stock return correlation during stress
- The main independent variable is quarterly ownership overlap between firm pairs
- The regression framework tests whether firms that are more closely connected in the ownership network exhibit stronger comovement during market stress